

**#This code will be used to demonstrate theorem 6.17. We will demonstrate that  $QI = \frac{x^2}{s^2}$  is decreasing along the  $\gamma I$  curve; (Part 3 — calculating the sign of the derivative of  $QI'(\gamma I(1), 1)$ )**

#Define QI;

$$QI := \frac{x^2 \cdot ((z + w)^2 - y^2)}{y^2 \cdot z \cdot w} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \left. \right\} :$$

$qI := \text{simplify}(\text{expand}(\text{subs}(\text{parametrization}, QI)))$  ;

$$qI := \frac{4 (a b + 1)^2 (-b + a)^2}{(a^2 + 1) (b^2 + 1) (a b - 1)^2} \quad (1)$$

#Define rI

$$\begin{aligned} rI := & \left( a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \right. \\ & + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\ & - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\ & - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \\ & - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\ & + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\ & + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\ & + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\ & - 4 b^{14}) a^2 - 2560 \left( b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\ & \left. - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \right) \end{aligned}$$

$$\begin{aligned}
& + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\
& + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\
& + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
& + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\
& - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\
& + 420 b^2 - 127) a^{20} + 8 \left( b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\
& + 2369 b^6 - 699 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left( b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\
& - 528 b^4 + \frac{4168}{9} b^2 + 1 \left. \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left( b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\
& - 1 \left. \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left( b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\
& - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left( b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \\
& + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left( b^8 - \frac{52}{7} b^6 \right. \\
& + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \left. \right) (b^2 + 1)^4 a^{19} + 5632 \left( b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 \\
& + 1)^4 a^{21} \Big) :
\end{aligned}$$

#Calculating the derivative of  $Q1$  at  $b=1$

$$d := \text{eval}(\text{diff}(q1, a), b = 1) \cdot \left( -\frac{\text{eval}(\text{diff}(r1, b), b = 1)}{\text{eval}(\text{diff}(r1, a), b = 1)} \right) + \text{eval}(\text{diff}(q1, b), b = 1) :$$

#Locate a range for  $\gamma l(1)$

$$\text{sturm}(\text{sturmseq}(\text{eval}(r1, b = 1), a), a, 2.6, 3); \text{\#number of roots in } [2.6, 3)$$

1

(2)

#Define  $nd$

$$nd := \text{numer}(d)$$

$$\begin{aligned} nd := & -2(a+1)(a^{35} + 5a^{34} + 4a^{33} + 5a^{32} - 6a^{31} - 34a^{30} - 238a^{29} - 1314a^{28} + 46a^{27} \\ & + 24066a^{26} + 120866a^{25} + 113794a^{24} - 131182a^{23} - 1404090a^{22} - 2400838a^{21} \\ & - 4220986a^{20} - 4431360a^{19} - 4976948a^{18} - 4426798a^{17} - 3600436a^{16} - 3237042a^{15} \\ & - 1751686a^{14} - 1442186a^{13} - 499334a^{12} - 523246a^{11} - 145874a^{10} - 41594a^9 \\ & - 49490a^8 - 218a^7 - 2974a^6 - 1218a^5 + 226a^4 - 65a^3 - a^2 + 2a - 1) \end{aligned} \quad (3)$$

#Define  $dd$

$$dd := \text{denom}(d)$$

$$\begin{aligned} dd := & (a^2 + 1)^2 a^3 (a^{28} - 7a^{24} - 208a^{22} + 405a^{20} + 23552a^{19} + 1232a^{18} + 10752a^{17} \\ & - 319075a^{16} - 48640a^{15} - 592416a^{14} - 26112a^{13} - 570845a^{12} + 23040a^{11} - 398944a^{10} \\ & + 19968a^9 - 143253a^8 + 5632a^7 - 42000a^6 - 4608a^5 + 647a^4 - 3584a^3 + 144a^2 \\ & - 65)(-1 + a) \end{aligned} \quad (4)$$

#Verify that  $nd$  and  $dd$  have the same sign in the interval.

$$\text{sturm}(\text{sturmseq}(nd, a), a, 2.6, 3); \text{\#number of roots in } [2.6, 3) ;$$

$$\text{sturm}(\text{sturmseq}(dd, a), a, 2.6, 3); \text{\#number of roots in } [2.6, 3)$$

0

0

(5)

#Calculate the sign of  $Q1'(\gamma l(1), 1)$  using  $Q1'(3, 1)$

$$\frac{\text{eval}(nd, a = 3)}{\text{eval}(dd, a = 3)}$$

$$-\frac{6096895832272}{564267245625}$$

(6)